The Four-Step Augmentation Pipeline and the Exact-Corruption Variant

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1 Where this sits in the pipeline

`EEGDataset.__getitem__` (mode `"ssl"`) reads *one* random 10 s window $x \in \mathbb{R}^{19\times 2000}$ (19 channels, 200 Hz) from a random recording, z-scores it per channel, then calls `_augment(x)` *twice independently* to produce the two views consumed by the SSL objective:

$$v_1 = \texttt{\_augment}(x), \qquad v_2 = \texttt{\_augment}(x).$$

Neither v_1 nor v_2 is the clean x — both are corrupted, each with its own draw of random numbers. The encoder never sees x itself; it only has to learn that v_1 and v_2 describe "the same thing" despite their independent corruptions.

2 The four steps inside `_augment`

Applied in this fixed order, each gated by its own coefficient:

1. **Amplitude scale jitter.** Per channel c, draw $s_c \sim \mathcal{U}(1-j,\, 1+j)$ with $j = \texttt{aug\_scale\_jitter} = 0.2$, then $x_c \leftarrow s_c \cdot x_c$.
2. **Additive Gaussian noise.** $x \leftarrow x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2)$, $\sigma = \texttt{aug\_noise\_std} = 0.1$ (z-scored units).
3. **Channel dropout.** Each of the 19 channels is independently zeroed with probability $p = \texttt{aug\_chan\_drop\_p} = 0.2$: $x_c \leftarrow x_c \cdot \mathbb{1}[u_c > p],\ u_c \sim \mathcal{U}(0, 1)$.
4. **Time masking — legacy vs. exact corruption.** Two mutually exclusive schemes, selected by `aug_exact_corruption`:

   Legacy (`aug_exact_corruption=False`, default). Zero out *one* random contiguous span of length $\ell \sim \mathcal{U}(0,\, f)\cdot T$, $f = \texttt{aug\_time\_mask\_frac} = 0.2$, at a random start position $s \sim \mathcal{U}(0,\, T-\ell)$. At most 20% of the window is touched, in a single block, identical across channels.

   Exact corruption (`aug_exact_corruption=True`). Per channel, independently: pick a random permutation of the $T = 2000$ time indices, then
   - zero-mask a fixed fraction $m = \texttt{aug\_mask\_frac} = 0.2$ of indices,
   - replace a *disjoint* fraction $o = \texttt{aug\_outlier\_frac} = 0.2$ of indices with outlier spikes $\pm k$ (in z-scored σ units, $k = \texttt{aug\_outlier\_scale} = 6.0$, random sign per spike).

   Exactly $m + o = 40\%$ of each channel's samples are corrupted, scattered (not contiguous), and independently chosen per channel.

3 What changes between the two schemes

Steps 1–3 (jitter, noise, channel dropout) are identical in both schemes — only step 4 differs:

	Legacy	**Exact corruption**
Corrupted fraction	up to 20%, one block	exactly 40%
Spatial pattern	contiguous span	scattered indices
Per-channel independence	same span, all channels	independent per channel
Corruption type	zero only	zero *and* $\pm 6\sigma$ outliers

Both schemes act on the *same* underlying x at the *same* point in the pipeline (after jitter/noise/dropout, before returning v_1/v_2) — exact corruption is not a new data source or an enlarged dataset, it is a stronger, differently-shaped corruption applied to the one random window already drawn.